Astral Classification Project

1. Overview

This project applies unsupervised machine learning methods (K-Means, HDBSCAN, and GMM) to cluster astronomical objects (STAR, GALAXY, QSO) using photometric data.

2. Setup Instructions

<u>Install dependencies</u>

Run the following:
pip install -r code/requirements.txt

3. Dataset

- Load the CSV file “star-galaxy-quasar.csv” from that data folder

4. Running the Project

Run notebooks in the following order:

1. code/01_Data_Cleaning_EDA.ipynb
2. code/02_KMeans.ipynb
3. code/03_GMM_Clustering.ipynb
4. code/04_HDBScan.ipynb

6. Models

Trained models are saved in: model/

- kmeans_model.pkl
- gmm_model.pkl
- hdbscan_model.pkl

5. Output

After running the notebook, the following outputs are generated.

- Performance metrics: Adjusted Rand Index (ARI) and Normalised Mutual Information (NMI) for K-means, GMM, and HDBSCAN, printed in each clustering's notebook.
- Contingency tables / matrix: Raw and row-normalised confusion matrices (as heatmaps and printed tables) showing the distribution of true classes (STAR, GALAXY, QSO) across predicted clusters.
- Figures – All plots (elbow curve, PCA projections) are displayed

HDBSCAN achieved the best performance among the 3 tested models.

6. Notes

- PCA is applied for dimensionality reduction while retaining 90% data variance (6 components)
- Stratified sampling is used for visualization

7. Authors

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